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Transient Thermal Analysis of Active and Passive Interventions for Next-Generation Wildland Firefighter Emergency Shelters

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Topic: Undergraduate Expo

Chris Schaefer, Ayden Wells, Jaylen Patel, Kristian Nasi, Ana Gegetchkori, Francisca Gomez Weiss
Department of Aerospace and Mechanical Engineering, Saint Louis University, St. Louis, MO, USA

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Project Objectives and Goals

The objective of this project is to enhance the safety and thermal performance of emergency fire shelters by investigating passive phase change material (PCM) and active cooling interventions using transient lumped-parameter modeling.

Background

Over the past 20 years, wildfires have increased in size, acreage burned, and intensity. In 2026, **112.3 million hectares burned globally** [1], and U.S. firefighter entrapment incidents reached a **20-year high** [2].

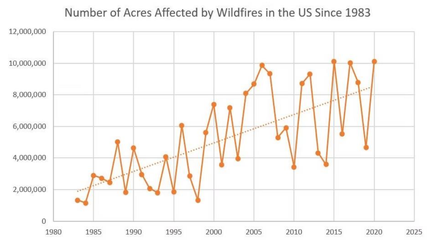


Figure 1: Data from NIFC. North Carolina state lands are exempt from 2004 fires and acreage [2]

Wildland fire burn overs account for over 20% of all wildland firefighter fatalities [14]

Inside the Shelter

The M2002 fire shelter protects firefighters from two types of heat:

Radiant Heat
- Travels in straight line
- Turns into heat when it contacts a cooler surface
- Shelter reflects 95%

Convective Heat
- Requires air movement
- Blasts of hot air transfers heat to surface
- Shelter absorbs 100%

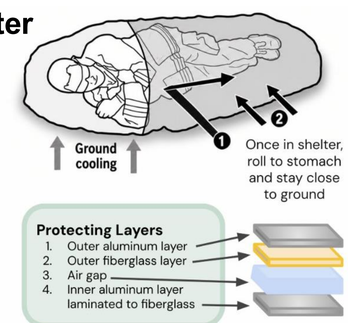


Figure 2: Emergency fire shelter protection mechanisms and material layers [13]

How They Work

Only to be used as a last resort, fire shelters offer firefighters protection in an emergency.

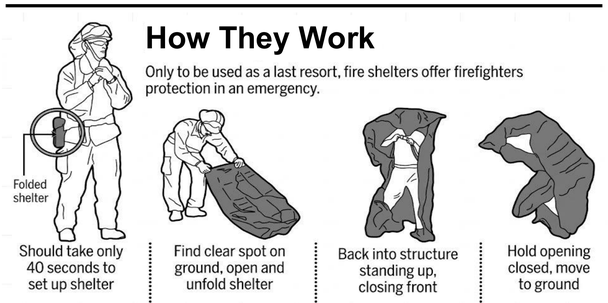


Figure 3: Emergency fire deployment procedure [13]

Methods & Procedures

Three phase-change materials were selected for transient thermal evaluation within the M-2002 fire shelter architecture.

- **RT42 Paraffin (Organic)**
- **PlusICE S46 (Inorganic)**
- **SP50 Gel (Inorganic)**

Property	RT42	S46	SP50 Gel
Type	Organic	Salt Hydrate	Inorganic
Melting Temp. (°C)	42	46	49–50
Latent Heat (kJ/kg)	165	110	220
Solid Density (kg/m³)	830	1610	1500
Liquid Density (kg/m³)	760	–	1400
Conductivity (W/mK)	0.20	0.60	0.60

Table 1: Material Properties of Selected PCMs [8]

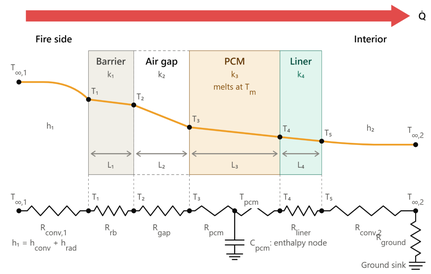


Figure 4: Thermal Resistance Network [12]

The baseline M-2002 shelter is modeled as a 6-node system; the PCM-integrated shelter utilizes a 7-node transient network. Each layer acts as a lumped node connected by thermal resistances and solved via MATLAB ODEs.

- **Energy Balance**:
$$m_i c_i \frac{dT_i}{dt} = Q_{in,i} - Q_{out,i}$$

- **PCM Node**:
$$\frac{dE_{PCM_5}}{dt} = Q_{45} - Q_{56} \Rightarrow T = f(E_{PCM}) \Rightarrow T_5 = T_{PCM,mp} + \frac{E_{PCM} - Q_{latent}}{mc_{p_5}}$$

- **Total Shelter Resistance**:
$$R_{total} = \frac{1}{h_1 A} + \sum \frac{L_i}{k_i A} + \frac{1}{h_2 A}$$

- **Heat Loading**:
$$Q = h_{conv} A(T_\infty - T_1) + \varepsilon\sigma A\left(T_{fire}^4 - T_1^4\right)$$

- **Heat Sink**:
$$\frac{dT_7}{dt} = \frac{(Q_{67} - Q_{ground})}{mc_{p_7}}$$

Data and Results

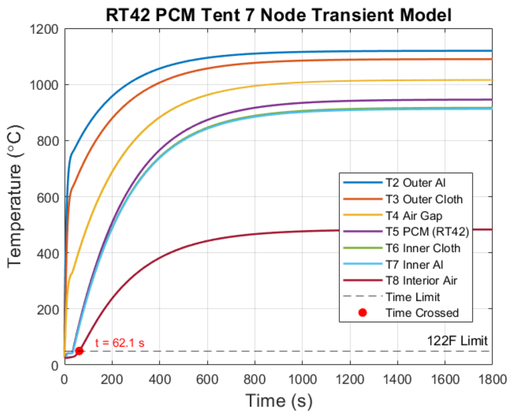


Figure 5: RT42 PCM Transient Model

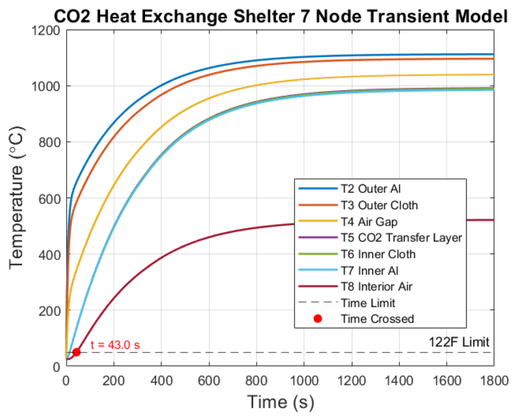


Figure 6: CO2 Heat Exchanger Transient Model

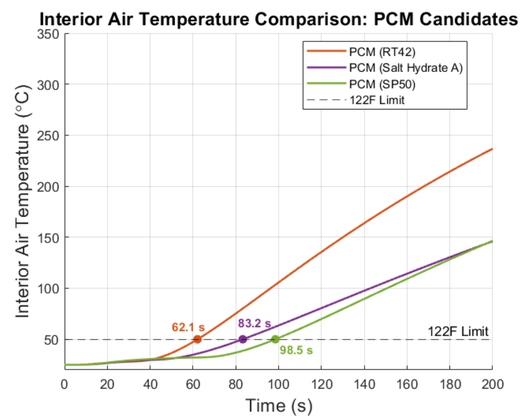


Figure 7: PCM Candidates Comparison

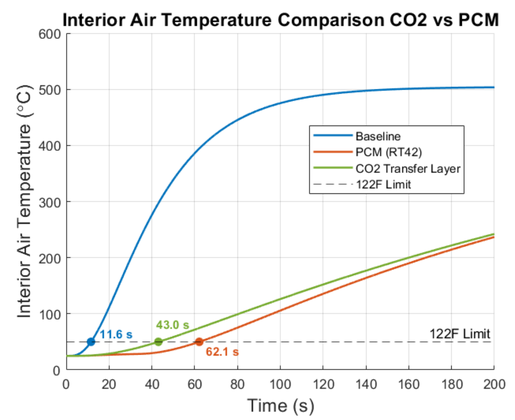


Figure 8: CO2 vs PCM Comparison

Conclusions

Material	Time (s)	Additional Mass (kg)	Improvement
Baseline	11.6	–	–
RT42 Paraffin	62.1	4.663	5.4×
PlusICE S46	83.2	9.046	7.2×
SP50 Gel	98.5	8.428	8.5×

- Both the passive PCM fire shelter and active CO_2 heat exchanger interventions extended the time before the shelter interior reaches 122°F compared to the baseline M-2002 shelter.
- The Paraffin RT42 PCM configuration provides the most effective temperature delay (62.1 s) to mass (4.66 kg) tradeoff.
- The CO_2 heat exchanger system provides active cooling, delaying the temperature limit to 39.1 s with a 1.10 kW peak heat removal rate; however, it is limited by its available cooling capacity and supply duration.

Future Studies / Recommendations

- Optimize PCM geometry, thickness, and placement
- Maximize thermal protection and minimize additional weight to the shelters
- Optimize tube arrangements and flow rates for CO_2
- Conduct laboratory and field experiments for data accuracy

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